

π from a Sliding Ruler on a Parabola

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Abstract

We present an elementary geometric construction that yields π from the antiderivative of $f(x) = x$. A rigid tangent segment of Euclidean length 1 is placed on each side of the contact point on the parabola $F(x) = x^2/2$ and swept from $-N$ to $+N$. The area of the region traced by this sliding ruler converges to π as $N \rightarrow \infty$. The proof requires only the Fundamental Theorem of Calculus and the standard integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$. The construction provides a new visual link between the identity function, its antiderivative, and the circle constant.

1 Introduction

The identity function $f(x) = x$ is the simplest non-constant polynomial. Its antiderivative $F(x) = x^2/2$ is the standard parabola. While the Fundamental Theorem of Calculus tells us that

$$\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}, \quad (1)$$

the antiderivative F does not directly display the area it encodes: one must evaluate F at two discrete points and subtract. This well-known fact motivates a different question: rather than extracting area from the *values* of F , can we extract geometric information from the *tangent lines* of F ?

We show that equipping the parabola with a rigid tangent of the simplest possible specification—unit length in each direction from the contact point—and sweeping it over the full curve produces a region whose area is exactly π .

2 Setup: from $f(x) = x$ to $F(x) = x^2/2$

We begin with the identity function and its indefinite integral:

$$f(x) = x, \quad F(x) = \int x \, dx = \frac{x^2}{2} + C. \quad (2)$$

At every point x_0 , the derivative of F recovers the original function regardless of C :

$$F'(x_0) = f(x_0) = x_0. \quad (3)$$

Since the tangent line depends only on F' , not on C , we may set $C = 0$ without loss of generality. The tangent to F at the contact point $(x_0, x_0^2/2)$ is therefore

$$y = x_0(x - x_0) + \frac{x_0^2}{2}, \quad (4)$$

with slope equal to x_0 itself—the value of the original function f .

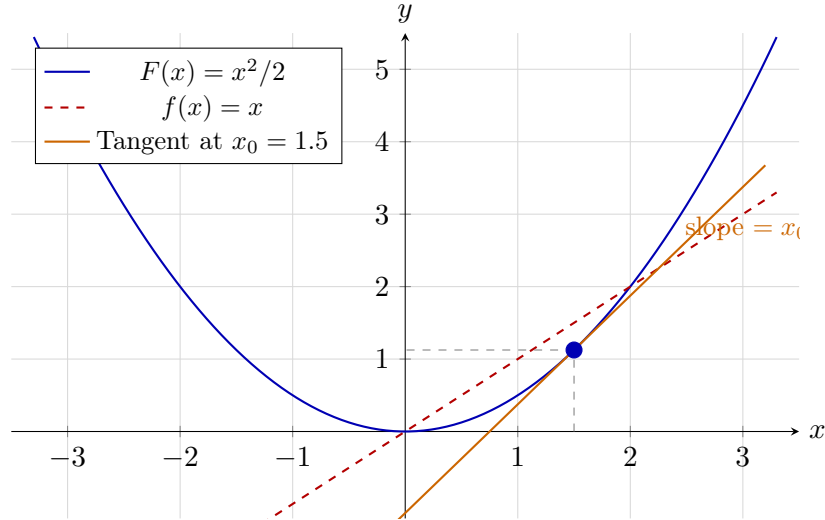


Figure 1: The parabola $F(x) = x^2/2$ with its tangent at $x_0 = 1.5$. The slope of the tangent equals $F'(x_0) = f(x_0) = x_0$: the derivative of the antiderivative returns the original function.

3 The Construction

Construction 1 (Sliding Ruler). At each contact point $x_0 \in \mathbb{R}$ on the parabola $F(x) = x^2/2$:

- (i) Compute the unit tangent direction: $\hat{t}(x_0) = \frac{1}{\sqrt{1+x_0^2}}(1, x_0)$.
- (ii) Place a rigid segment of Euclidean length 1 in the positive tangent direction and length 1 in the negative tangent direction.
- (iii) The total ruler has length 2, centred at the contact point.

As x_0 varies over $[-N, N]$, the union of all such segments forms the *tangent sweep* $\mathcal{S}_N$. The *swept area* is $\mathcal{A}(N) = \text{Area}(\mathcal{S}_N)$.

The endpoints of the ruler at contact point x_0 are:

$$P_{\pm}(x_0) = \left(x_0 \pm \frac{1}{\sqrt{1+x_0^2}}, \frac{x_0^2}{2} \pm \frac{x_0}{\sqrt{1+x_0^2}} \right). \quad (5)$$

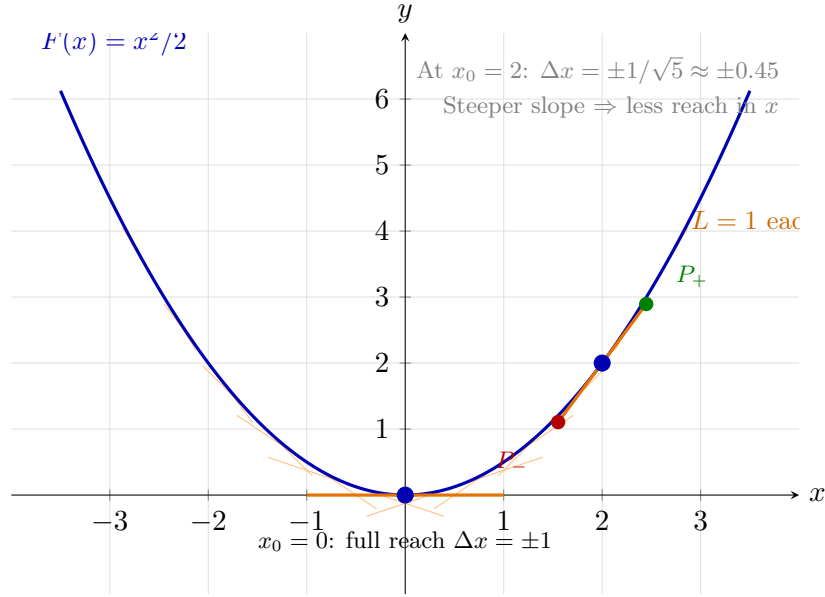


Figure 2: The sliding ruler construction. At $x_0 = 0$ the tangent is horizontal and the ruler extends ± 1 in x . At $x_0 = 2$ the slope steepens and the x -reach compresses to $\pm 1/\sqrt{5}$, while the Euclidean length remains 1 in each direction.

4 The Swept Area

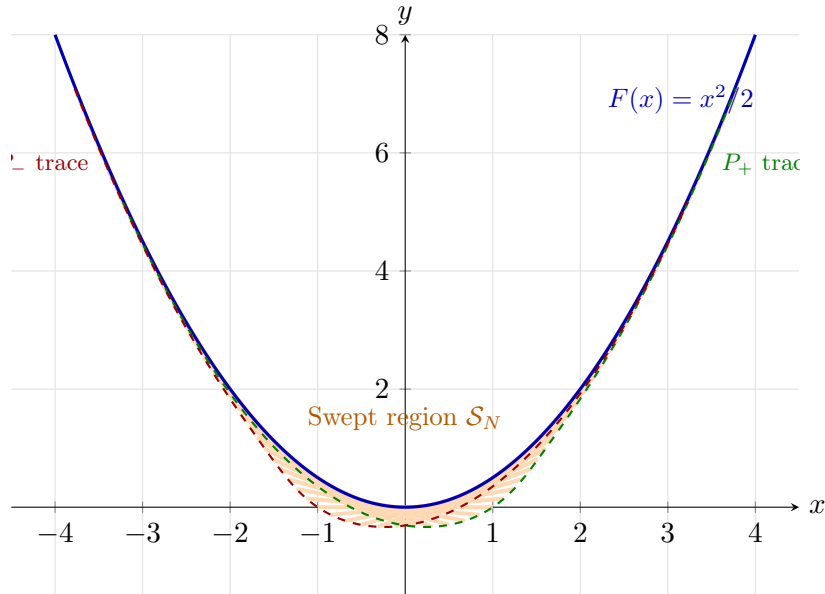


Figure 3: The swept region $\mathcal{S}_N$ for $N = 3.5$, formed by the union of all fixed-length tangent segments. The dashed curves show the traces of the ruler tips P_+ and P_- . The ribbon narrows as the slope increases because the x -reach compresses while the Euclidean length stays fixed.

At each contact point x_0 , the tangent line (4) at a general x -value gives

$$y_{\text{tan}}(x; x_0) = x_0 x - \frac{x_0^2}{2}. \quad (6)$$

The curve itself has $F(x) = x^2/2$, so the *vertical gap* between the parabola and its tangent at horizontal position x is

$$F(x) - y_{\tan}(x; x_0) = \frac{x^2}{2} - x_0 x + \frac{x_0^2}{2} = \frac{(x - x_0)^2}{2}. \quad (7)$$

This gap is always non-negative: the tangent lies *below* the parabola (since F is convex), and the gap equals zero only at the contact point.

For the fixed-length ruler, the tips reach $x = x_0 \pm 1/\sqrt{1 + x_0^2}$, so the maximum gap at each tip is

$$\frac{1}{2} \cdot \frac{1}{1 + x_0^2}. \quad (8)$$

5 Main Result

Theorem 2. *Let $\mathcal{A}(N)$ denote the area swept by the construction of Section 3 as x_0 ranges over $[-N, N]$. Then*

$$\boxed{\lim_{N \rightarrow \infty} \mathcal{A}(N) = \pi} \quad (9)$$

Proof. As x_0 advances by an infinitesimal dx_0 , the tangent direction rotates by an angle

$$d\theta = \frac{d}{dx_0} \arctan(x_0) dx_0 = \frac{dx_0}{1 + x_0^2}. \quad (10)$$

A rigid segment of length L rotating through an infinitesimal angle $d\theta$ sweeps an area $\frac{1}{2}L^2 d\theta$ (the area of an infinitesimal circular sector of radius L). Our ruler has two arms, each of length 1, so the total infinitesimal swept area is

$$dA = 2 \cdot \frac{1}{2} \cdot 1^2 \cdot d\theta = \frac{dx_0}{1 + x_0^2}. \quad (11)$$

Integrating over the full sweep:

$$\mathcal{A}(N) = \int_{-N}^N \frac{dx_0}{1 + x_0^2} = \arctan(N) - \arctan(-N) = 2 \arctan(N). \quad (12)$$

Taking the limit:

$$\lim_{N \rightarrow \infty} \mathcal{A}(N) = 2 \lim_{N \rightarrow \infty} \arctan(N) = 2 \cdot \frac{\pi}{2} = \pi. \quad (13)$$

□

Remark 3 (Non-overlapping sectors). *Equivalently, the sweep can be parameterised by the tangent angle $\theta = \arctan(x_0) \in (-\pi/2, \pi/2)$ rather than by x_0 . In this parameterisation the construction becomes a continuous rotation of a rigid segment through a total angle π , and each infinitesimal increment $d\theta$ produces a circular sector that does not overlap with the preceding one. The swept area is therefore*

$$\mathcal{A} = \int_{-\pi/2}^{\pi/2} d\theta = \pi, \quad (14)$$

with no need to account for overlapping tangent segments in the x_0 -parameterisation. This confirms that the integral $\int (1 + x_0^2)^{-1} dx_0$ computes the geometric area of the swept region exactly.

6 Injectivity of the Tangent Sweep

The proof in Theorem 2 and Remark 3 assumes that the infinitesimal sectors do not overlap, so that integrating $d\theta$ faithfully counts the geometric area. This section makes the non-overlapping condition precise.

6.1 Parameterisation of the swept region

A general point on the ruler at contact point x_0 is described by the map

$$\mathbf{r}(x_0, s) = \left(x_0 + \frac{s}{\sqrt{1+x_0^2}}, \frac{x_0^2}{2} + \frac{s x_0}{\sqrt{1+x_0^2}} \right), \quad s \in [-1, 1], \quad (15)$$

where the contact point is $(x_0, x_0^2/2)$, the unit tangent direction is $\hat{t}(x_0) = (1+x_0^2)^{-1/2}(1, x_0)$, and s measures signed Euclidean distance along the tangent. The tips of the ruler correspond to $s = \pm 1$.

For the swept area to equal π exactly, the mapping $(x_0, s) \mapsto \mathbf{r}(x_0, s)$ must be injective (one-to-one) on its domain: if two distinct tangent segments overlapped within their unit lengths, the integral would double-count area and overstate the physical region.

6.2 The radius-of-curvature criterion

In differential geometry, the family of tangent lines to a plane curve begins to self-intersect at the curve's *evolute*—the locus of centres of curvature [4, 5]. A tangent segment of half-length L from the curve remains non-overlapping provided

$$L \leq R_{\min} = \min_{x_0} R(x_0), \quad (16)$$

where $R(x_0)$ is the radius of curvature at x_0 . This is a consequence of the tubular-neighbourhood theorem for plane curves [4, 6].

For the parabola $F(x) = x^2/2$:

$$\begin{aligned} F'(x) &= x, & F''(x) &= 1, \\ R(x_0) &= \frac{(1 + (F')^2)^{3/2}}{|F''|} = (1 + x_0^2)^{3/2}. \end{aligned} \quad (17)$$

Since $x_0^2 \geq 0$, the minimum radius of curvature occurs at the vertex:

$$R_{\min} = R(0) = 1. \quad (18)$$

Our ruler half-length is $L = 1$, so the criterion (16) is satisfied with equality at the vertex and strictly at every other point.

Remark 4 (Boundary degeneracy at the vertex). *At $x_0 = 0$ the ruler tips land exactly on the evolute of the parabola (whose cusp sits at $(0, 1)$). The sweep map (15) is injective on each open half $\{s > 0\}$ and $\{s < 0\}$ separately; the two halves meet along the parabola itself ($s = 0$), which is a set of measure zero and does not affect the area integral. At the vertex, however, $L = R_{\min}$, so the construction is at the exact boundary of the non-overlap criterion. Adjacent tangent segments from nearby contact points do not cross within $|s| < 1$, but extending the ruler beyond unit length would cause overlap near $x_0 = 0$ [4].*

6.3 Jacobian verification

The Jacobian determinant of $\mathbf{r}$ with respect to (x_0, s) provides a direct, computational confirmation of the area. We compute each partial derivative of (15). Writing $\sigma(x_0) = (1+x_0^2)^{1/2}$ for brevity:

$$\begin{aligned} \frac{\partial x}{\partial x_0} &= 1 - \frac{s x_0}{\sigma^3}, & \frac{\partial x}{\partial s} &= \frac{1}{\sigma}, \\ \frac{\partial y}{\partial x_0} &= x_0 + \frac{s}{\sigma} - \frac{s x_0^2}{\sigma^3} = x_0 + \frac{s}{\sigma^3}, & \frac{\partial y}{\partial s} &= \frac{x_0}{\sigma}. \end{aligned} \quad (19)$$

The Jacobian determinant is

$$\begin{aligned}
J(x_0, s) &= \frac{\partial x}{\partial x_0} \frac{\partial y}{\partial s} - \frac{\partial x}{\partial s} \frac{\partial y}{\partial x_0} \\
&= \left(1 - \frac{s x_0}{\sigma^3}\right) \frac{x_0}{\sigma} - \frac{1}{\sigma} \left(x_0 + \frac{s}{\sigma^3}\right) \\
&= \frac{x_0}{\sigma} - \frac{s x_0^2}{\sigma^4} - \frac{x_0}{\sigma} - \frac{s}{\sigma^4} \\
&= -\frac{s(1 + x_0^2)}{\sigma^4} = -\frac{s}{(1 + x_0^2)}.
\end{aligned} \tag{20}$$

Two features are immediate:

- (i) J is *antisymmetric* in s : the map reverses orientation as the ruler crosses the contact point. Geometrically, the parabola itself ($s = 0$) is a fold line of the sweep.
- (ii) J vanishes on $s = 0$ (the curve) and nowhere else for finite x_0 and $s \neq 0$.

The geometric area of the swept region is given by the integral of $|J|$:

$$\mathcal{A} = \int_{-\infty}^{\infty} \int_{-1}^1 |J(x_0, s)| ds dx_0 = \int_{-\infty}^{\infty} \int_{-1}^1 \frac{|s|}{1 + x_0^2} ds dx_0. \tag{21}$$

The inner integral evaluates to

$$\int_{-1}^1 \frac{|s|}{1 + x_0^2} ds = \frac{2}{1 + x_0^2} \int_0^1 s ds = \frac{2}{1 + x_0^2} \cdot \frac{1}{2} = \frac{1}{1 + x_0^2}, \tag{22}$$

so that

$$\mathcal{A} = \int_{-\infty}^{\infty} \frac{dx_0}{1 + x_0^2} = \pi, \tag{23}$$

in exact agreement with (11) and the sector argument. Each arm of the ruler ($s > 0$ and $s < 0$) contributes $\pi/2$ independently, confirming the “two half-sectors” reasoning of the proof of Theorem 2.

Remark 5 (Generality of the criterion). *For a ruler of half-length ℓ on any smooth convex curve, the sweep is non-overlapping whenever $\ell \leq R_{\min}$. For the parabola $F(x) = ax^2/2$, one has $R_{\min} = 1/a$, so the critical ruler half-length is $\ell = 1/a$. Choosing $a = 1$ and $\ell = 1$ is the unique setting in which the non-overlap condition is saturated and the swept area equals exactly π .*

7 Convergence

The convergence is illustrated in Table 1 and Figure 4. The rate is

$$\pi - \mathcal{A}(N) = \pi - 2 \arctan(N) \sim \frac{2}{N} \quad \text{as } N \rightarrow \infty, \tag{24}$$

which is slow (order $1/N$), but the convergence is monotone and from below.

N	$\mathcal{A}(N) = 2 \arctan(N)$	$\pi - \mathcal{A}(N)$
1	1.570 796 ...	1.571
5	2.746 801 ...	3.95×10^{-1}
10	2.942 200 ...	1.99×10^{-1}
50	3.101 593 ...	4.00×10^{-2}
100	3.121 593 ...	2.00×10^{-2}
1 000	3.139 593 ...	2.00×10^{-3}
10^6	3.141 591 ...	2.00×10^{-6}

Table 1: Convergence of $\mathcal{A}(N) = 2 \arctan(N)$ to π .

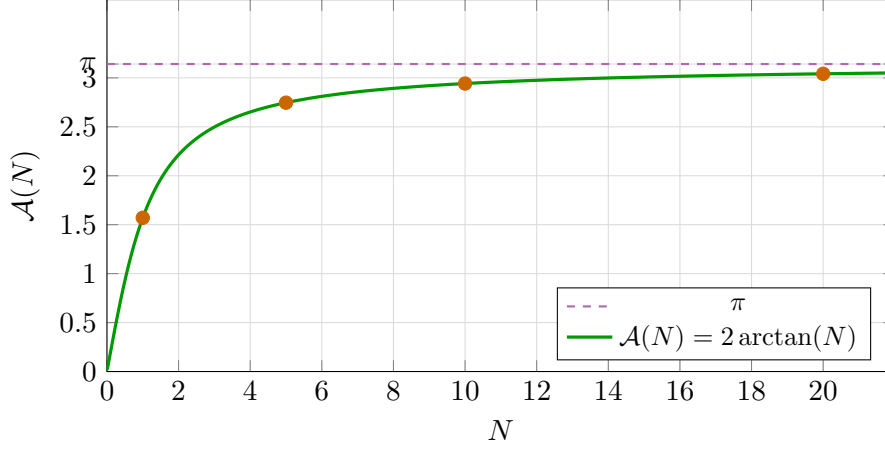


Figure 4: The swept area $\mathcal{A}(N)$ as a function of the sweep bound N , converging monotonically to π from below.

8 Why π : the Total Turning Angle

The appearance of π has a clean geometric explanation. As x_0 ranges from $-\infty$ to $+\infty$, the tangent angle

$$\theta(x_0) = \arctan(x_0) \quad (25)$$

increases from $-\pi/2$ to $+\pi/2$. The *total turning angle* of the parabola is therefore

$$\Delta\theta = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi. \quad (26)$$

A rigid unit segment sweeping through angle $d\theta$ contributes area $d\theta$ (from two half-sectors of radius 1). Integrating $d\theta$ over the full turning angle gives π .

The parabola opens to infinity, yet its tangent direction is bounded: it asymptotically approaches vertical but never reaches it. This *finite angular budget* is what forces the swept area to converge. The factor $1/(1+x_0^2)$ in the integrand is the derivative of $\arctan$, which is the same kernel that defines the Cauchy distribution in probability and appears in the arc-length parameterisation of the unit circle.

Remark 6. *The result holds for any parabola $F(x) = ax^2/2$ (with $a > 0$): the total turning angle is always π , independent of the coefficient. Rescaling a changes the curvature at each point but not the total angular range of the tangent.*

Remark 7. *If the ruler has length ℓ in each direction (total length 2ℓ) instead of 1, the swept area becomes $\ell^2 \pi$. The choice $\ell = 1$ —the smallest whole unit, placed symmetrically—is the unique specification that yields π without a prefactor.*

Remark 8 (The vanishing constant). *Integrating $f(x) = x$ gives $F(x) = x^2/2 + C$, and every calculus textbook warns: do not forget the constant. Yet the tangent slope at x_0 is $F'(x_0) = x_0$, regardless of C . The construction passes through integration (which introduces C) and immediately back through differentiation (which removes it): $-C + C = 0$. The constant of integration—the one quantity students are taught never to omit—is precisely the quantity that plays no role in the result. This is why the swept area is π for every vertical translate $x^2/2 + C$ of the parabola: the geometry of the tangent directions is invariant under vertical shifts.*

9 Connection to Prior Work

The underlying integral $\int_{-\infty}^{\infty} (1 + x^2)^{-1} dx = \pi$ is one of the most classical results in analysis, known since Leibniz and Gregory in the 1670s. The total turning angle of a parabola being π is standard in differential geometry. Mamikon Mnatsakanian’s sweeping-tangent theorem [2, 3] — which establishes that the area of a tangent sweep equals the area of its tangent cluster — has been applied to parabolic segments, exponentials, cycloids, and oval rings [1, 2], using both variable-length and constant-length tangent segments. None of the mathematics in this paper is new.

What may be new is the *presentation*: the specific construction of placing a rigid unit tangent on the antiderivative of $f(x) = x$ and observing that its swept area converges to π . This packages several well-known facts into a single visual construction that, to the author’s knowledge, has not appeared in this form before. The author makes no stronger claim than that.

The present construction differs from the Apostol–Mnatsakanian framework in two respects:

- (i) The curve is *open* (a complete parabola), so the total turning angle is π rather than 2π .
- (ii) The tangent length is fixed at 1 in Euclidean measure on each side, which converts the turning angle directly into area via $\mathcal{A} = \int d\theta = \pi$.

10 Summary

The construction is:

Rule. Integrate $f(x) = x$ to obtain $F(x) = x^2/2$. Choose a tangent of total length 2 (one unit in each direction from the contact point). Sweep it from $-\infty$ to $+\infty$. The area it traces is π .

Starting from the simplest non-constant function, its antiderivative, and a rigid ruler of the smallest symmetric whole-unit length, the construction produces the circle constant π —with no circles anywhere in sight.

References

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